



Cycling profile of MgAl_2O_4 -incorporated composite electrolytes composed of PEO and LiPF_6 for lithium polymer batteries

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ABSTRACT

Magnesium aluminate (MgAl_2O_4)-incorporated poly(ethylene oxide) (PEO)–lithium hexafluorophosphate (LiPF_6) based composite polymer electrolyte (CPE) membranes were prepared by a hot press for the first time. The membranes were subjected to X-ray diffraction (XRD), scanning electron microscopy (SEM), thermogravimetric (TG), differential scanning calorimetry (DSC), tensile, impedance spectroscopy, compatibility and transport number studies. The incorporation of MgAl_2O_4 greatly enhanced the ionic conductivity, compatibility and mechanical integrity of the polymeric membrane. Finally, an all solid state lithium cell composed of Li/CPE/LiFePO_4 was assembled and its cycling profile was analyzed at 70 °C. The cells delivered a discharge capacity of 127 mAh g^{-1} at 1 C-rate with very good capacity retention up to 100 cycles which is found to be better than those reported earlier.

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1. Introduction

Rechargeable lithium batteries represent an excellent choice for electrochemical power sources characterized by high energy density, good cyclability and reliability [1]. The realization of such power is beset with several problems, especially concerning the electrolyte [2–4]. In this context, lithium polymer batteries have many advantages over their liquid counterparts such as high energy density, flexible geometry, no-leakage of electrolyte, thus higher safety [5]. Generally, solid polymer electrolytes (SPEs) offer very low ionic conductivity of the order of $10^{-8} \text{ S cm}^{-1}$ at ambient temperature that excludes their practical applications [6]. On the other hand, gel–polymer electrolytes (GPEs) lose their mechanical strength when they are plasticized [7]. Several studies reveal that nanocomposite polymer electrolytes can lead to the assembly of safe and reliable batteries [8].

Poly(ethylene oxide), PEO-based SPEs have a series of specific features, such as low cost, good chemical stability and safety [9]. The crystalline to amorphous transition in the $\text{PEO:LiX-Al}_2\text{O}_3$

composite electrolyte system has been analyzed by DSC and conductivity measurements by Kumar and Rodrigues [10]. The same group has also analyzed the structure, evolution and conductivity of $\text{PEO:LiBF}_4\text{-MgO}$ composite electrolytes [11], demonstrating that the addition of MgO significantly suppressed the PEO melting temperature and retarded its crystallization kinetics.

In the present study, in order to exploit the advantages of using both ceramic fillers at the same time (i.e., MgO and Al_2O_3), MgAl_2O_4 was synthesized and incorporated at different percentages into PEO matrixes comprising LiPF_6 salt and their physical and electrochemical properties were thoroughly studied. Recent studies indicated that polymeric membranes prepared by conventional solvent casting method demonstrated poor interfacial properties at the lithium/polymer electrolyte interface [12]. Impurities, mostly the traces of solvent, are trapped in the high surface area nanosized inert fillers in the solvent cast electrolytes even after prolonged drying [13]. Hence in the present study, the hot press technique was employed for the preparation of composite polymer electrolytes (CPEs). Hydrothermally synthesized nanostructured lithium iron phosphate (LiFePO_4) was used as cathodic material for charge/discharge cycling studies in lithium cells as it was found to be the most suitable candidate for elevated temperature applications when composite polymer electrolytes are used [14].

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Table 1
Composition of PEO, MgAl₂O₄ and lithium salt in the different electrolyte membranes.

Sample	PEO (wt.%)	MgAl ₂ O ₄ (wt.%)	LiPF ₆ (wt.%)
S1	95	0	5
S2	90	5	5
S3	85	10	5
S4	75	17	8
S5	70	20	10
S6	94	5	1
S7	93	5	2
S8	92	5	3
S9	91	5	4

2. Experimental

2.1. Synthesis of magnesium aluminate (MgAl₂O₄) filler

The magnesium aluminate filler particles were prepared by sol–gel method using citric acid as a chelating agent. Stoichiometric mixtures of MgCl₂ and Al(NO₃)₃·9H₂O (99% purity Aldrich, USA) were dissolved in deionized water and added drop wise, under continuous stirring, to an agitated aqueous solution of citric acid. The metal to chelating agent ratio was fixed to be 1:2. Initially the pH (pH metre supplied by Metrohm, Italy) of the solution was around 2. The pH of the solution was adjusted in the range of 5.5–7 by using ammonium hydroxide. Then, the resultant solution was evaporated at 70–80 °C until a sol–gel was obtained. The resulting precursor was decomposed in air and finally calcined at 800 °C for 3 h [15]. All chemicals have been purchased from Aldrich, USA.

2.2. Preparation of composite polymer electrolyte (CPE) membranes

Before use, poly(ethylene oxide) (99% purity, Aldrich, USA) and lithium hexafluorophosphate (LiPF₆, 99% purity, Merck, Germany) were dried under vacuum for 48 h at 50 °C and 80 °C, respectively. The MgAl₂O₄ filler was also dried under vacuum at 90 °C for 96 h before use. Composite polymer electrolytes were prepared by dispersing appropriate amounts of filler in complexes of PEO and LiPF₆ as displayed in Table 1. The resultant powder was hot-pressed into films, as described elsewhere [16–19]. The polymer electrolyte films had an average thickness of 30–50 μm. This procedure yielded homogenous and mechanically strong membranes, which were dried under vacuum at 50 °C for 24 h for further characterization.

2.3. Physical characterization

X-ray diffraction (XRD) measurements were conducted on a Bruker D8 X-ray diffractometer (Cu Kα radiation, λ = 1.54178 Å) at 50 kV and 250 mA with a scanning rate of 2°/min.

Morphological examination of the films was made by a field emission scanning electron microscope (FESEM, model S-4700, Hitachi) under a vacuum condition (10^{−1} Pa) after sputtering gold on one side of the films.

Differential scanning calorimetry (Mettler Toledo, Switzerland) measurements were performed between −100 and +100 °C, while TG-DTA (TA Instruments, USA) in the temperature range of 30–400 °C, both at a rate of 10 °C min^{−1}.

The mechanical strength of the composite polymer electrolytes was determined using a tensile machine (Tinius Olsen) with a constant cross-head speed of 10 mm min^{−1}. The sample dimensions were 20 mm (W) × 50 mm (L) × 0.1 mm (T).

2.4. Electrochemical characterization

The ionic conductivity of the polymer electrolyte membranes sandwiched between two stainless steel blocking electrodes (10 mm diameter) was measured using an electrochemical impedance analyser (IM6-Bio Analytical Systems, Germany) in the 50 mHz to 100 kHz frequency range at various temperatures (i.e., 0, 15, 30, 45, 60 and 70 °C).

Symmetric non-blocking cells in which the selected polymer electrolyte was sandwiched between two lithium metal electrodes (Li/CPE/Li) were assembled for evaluating the interfacial stability; it was investigated by studying the time dependence of the impedance of the systems after prolonged storage under open-circuit potential at 50 °C.

The lithium transference number was calculated by the method proposed by Vincent and co-workers [20–22], thus adopting the following Eq. (1):

$$t_{\text{Li}}^+ = \frac{I_{\text{ss}}(V - I_0 R_0)}{I_0(V - I_{\text{ss}} R_{\text{ss}})} \quad (1)$$

In a typical experiment, a Li/CPE/Li cell was polarized by a dc pulse of 10 mV (IM6-Bio Analytical Systems, Germany). Time evolution of the resulting current flow was then followed. The initial (*I*₀) and steady state (*I*_{ss}) values of current flowing through the cell during the polarization were measured. *R*₀ and *R*_{ss} represent the resistance values before and after the perturbation of the system. Impedance spectra were made before and after the pulse application in order to correct the changes.

2.5. Electrode preparation, lithium polymer cell assembly and electrochemical tests

The LiFePO₄/C cathode material was synthesized in the form of nanostructured powder through a mild hydrothermal procedure described in details in Meligrana et al. [23]. The composite cathode was prepared in the form of a film (average thickness of about 70 μm) by blending 10 wt.% of poly(vinylidene fluoride) as the binder (SolvaySolef 6020) with 20 wt.% of acetylene black (Shawinigan Black AB50, Chevron Corp., USA) as the electronic conductivity enhancer and 70 wt.% of LiFePO₄/C active material, thoroughly mixed in 1-methyl-2-pyrrolidone (NMP, Aldrich). The paste was coated by a doctor-blade process onto an aluminium foil current collector. After the complete evaporation of the NMP solvent by mild heating and progressive vacuum treatment for 24 h, electrode disks of 2.54 cm² were punched out and dried under high vacuum at 130 °C for 5 h. The lithium polymer cell was fabricated by laminating the three components in sequence, i.e., a lithium metal anode foil, a layer of the CPE membrane and the LiFePO₄/C composite cathode film. The assembly was housed into a hermetically sealed test cell (model ECC-Std, <http://el-cell.com/products/testcells/ecc-std>, purchased from EL-Cell, Germany). All preparations were performed in an argon-filled glove box (MBraun Labstar, Germany) having a humidity content below 1 ppm.

The characteristics and performance of the all-solid lithium polymer cell were investigated at 70 °C in terms of charge/discharge galvanostatic cycling by an Arbin Instrument Testing System model BT-2000, setting the cut off voltages to 2.50–4.00 V vs. Li/Li⁺. The charge–discharge cycles were set at the same rate at different current rates.

3. Results and discussion

3.1. XRD analysis

Fig. 1 shows the X-ray spectra of PEO (a, black line), MgAl₂O₄ (b, green line), sample S1 made of PEO and LiPF₆ in the 95:05 wt.%

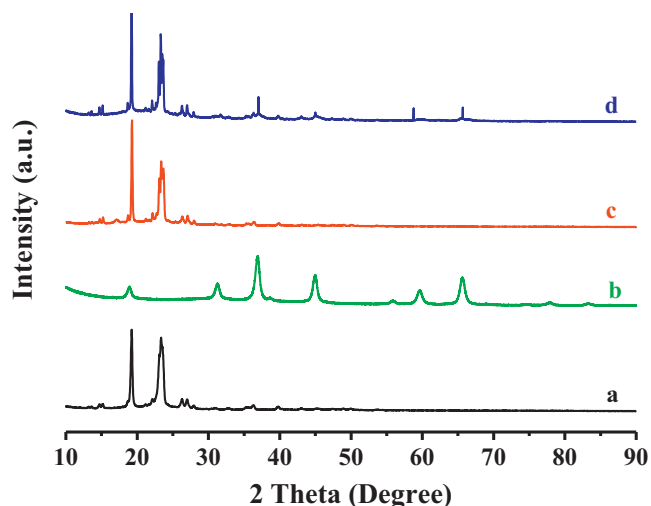


Fig. 1. XRD patterns of (a) PEO, (b) MgAl_2O_4 , (c) S1 (PEO+ LiPF_6) and (d) S3 (PEO+ MgAl_2O_4 + LiPF_6). (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.)

(c, red line) and sample S3 composed of PEO, MgAl_2O_4 and LiPF_6 in the 85:10:05 wt.% (d, blue line). The diffraction peaks at 31, 39, 45, 60 and 65 (2θ) in pattern (b) correspond to the (hkl) values (2 2 1), (3 1 1), (4 0 0), (4 2 2) and (4 4 0), respectively, indicating the crystalline nature of MgAl_2O_4 [24,25]. It is seen from pattern (c) that the intensity of PEO is reduced upon the incorporation of LiPF_6 salt. Further, the addition of MgAl_2O_4 (pattern d) shows a reduction in the intensity of the diffraction peaks, which indicates an increase in amorphous nature of the polymeric membrane. A

similar observation was reported by Benedict et al. [26] for XRD analysis of plasticized PEO-based gel-polymer electrolytes and Manuel Stephan et al. [27] for PVdF-HFP-based composite electrolytes.

3.2. SEM analysis

Fig. 2(a) and (b) shows the SEM images of PEO+ LiPF_6 (PEO: LiPF_6 : 95:05 wt.%) and PEO+ MgAl_2O_4 + LiPF_6 (PEO: MgAl_2O_4 : LiPF_6 : 85:10:05 wt.%) membranes. According to Chu et al. [28,29], the morphology of the electrolyte surface can be modified/tailored by the incorporation of both ionic salts and fillers. Fig. 2(a) and (b) displays a very rough and inhomogeneous surface morphology for the PEO+ LiPF_6 membrane. The incorporation of MgAl_2O_4 in the polymeric membrane facilitates the formation of a smooth surface morphology as seen particularly in Fig. 2(c). The smooth morphology is also attributed to a reduction in the crystallinity of PEO due to cross-linking with cations of both lithium ions and MgAl_2O_4 [13,29]. Nevertheless, some of the particles are agglomerated and formation of clusters is also observed at higher magnification (see image d).

3.3. Thermal studies

Lithium is a highly unstable metal at elevated temperatures and can react with the components of the electrolytes based on alkyl carbonate solvents, e.g., in batteries assembled with lithiated graphite anodes and delithiated oxide cathodes (e.g., Li_xCoO_2 , $x=0.5$) [30]. At elevated temperatures, the solid electrolyte interface (SEI) on the graphite anode gets destroyed allowing rapid and direct reaction with the lithiated graphite under the passivating layer [31]. In this respect, a fully solid polymer electrolyte plays

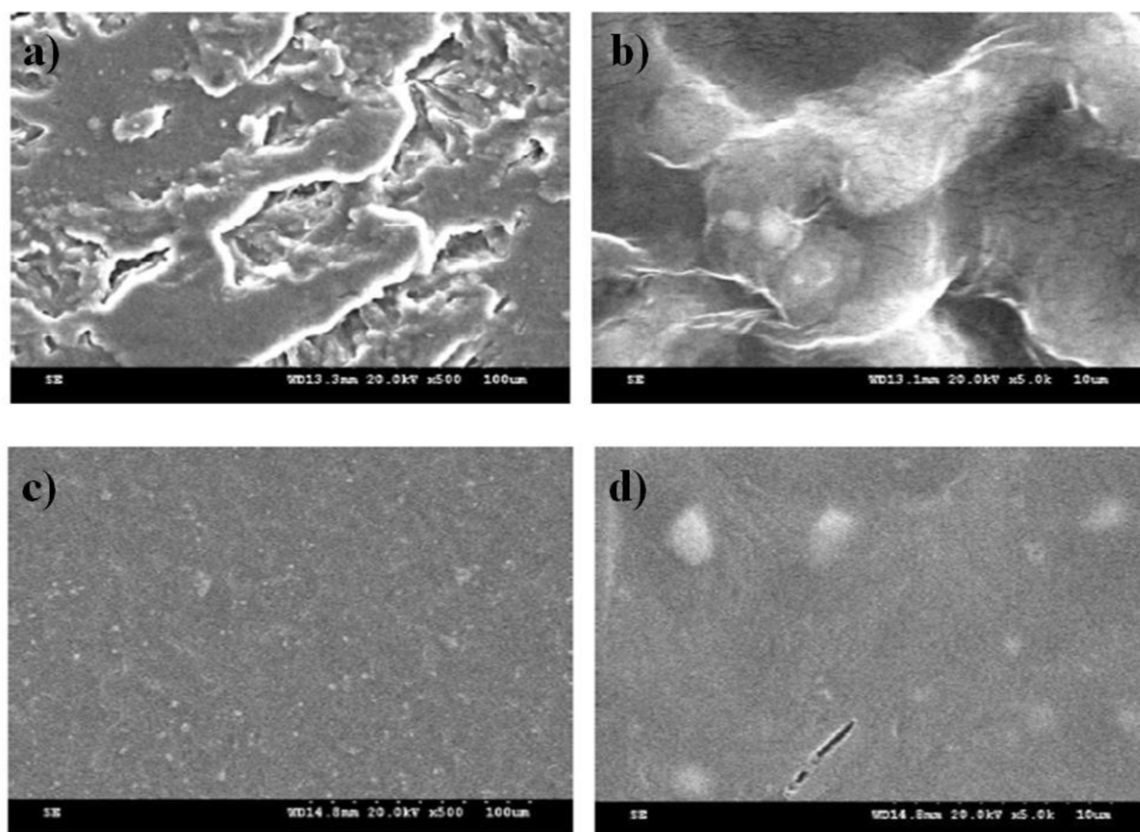


Fig. 2. SEM images: (a and b) S1 (PEO+ LiPF_6); (c and d) S3 (PEO+ MgAl_2O_4 + LiPF_6) with two different magnifications (i.e., 500 and 5000).

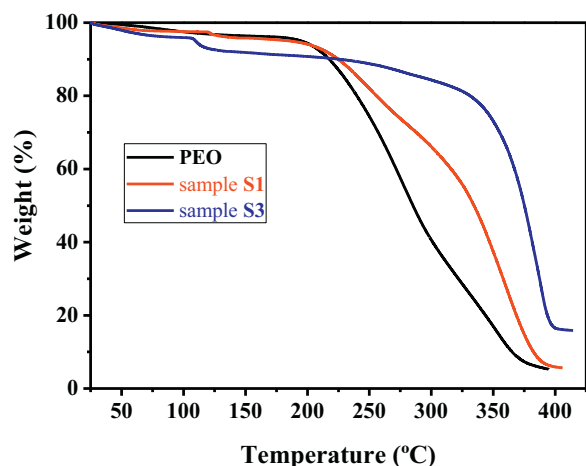


Fig. 3. Thermo-gravimetric analysis of PEO, S1 (PEO + LiPF₆) and S3 (PEO + MgAl₂O₄ + LiPF₆).

a vital role in the safety of lithium batteries. In order to ascertain the possibility of employing PEO-based electrolytes, the prepared membranes were subjected to TG and DSC analyses.

The TG traces of (a) PEO, (b) PEO + LiPF₆ (sample S1) and (c) PEO + MgAl₂O₄ + LiPF₆ (sample S3) are shown in Fig. 3. A negligible weight loss of around 3% is observed around at 40 °C in the case of PEO and is attributed to the removal of moisture absorbed at the time of loading the sample. Further, an endothermic peak is seen around at 65 °C and is attributed to the melting of PEO [32]. The weight loss is confined to a single event which occurs at around 200 °C and corresponds to the decomposition of PEO. These results are in accordance with those reported earlier for PEO-based electrolytes [32]. On the other hand, the thermal stability of LiPF₆ and LiPF₆ + MgAl₂O₄-added polymeric membranes (Fig. 3(b) and (c)) is found to be increased and the thermal stability increases to above 300 °C (% of wt. loss at 300 °C is about 9.5%). This is an indication of the fact that PEO + LiPF₆ + MgAl₂O₄ is stable up to a temperature of 330 °C in nitrogen atmosphere, which is fairly high when compared to the operating temperature of a practical lithium–polymer cell, viz. 60–70 °C. The increased thermal stability of MgAl₂O₄-added composite electrolytes may be attributed to the intercalation and exfoliation of the polymer matrix with inert particles, which results in a strong barrier effect preventing from the thermal degradation to a certain extent.

In the case of composite polymer electrolytes, the differential scanning calorimetric measurements are useful for the

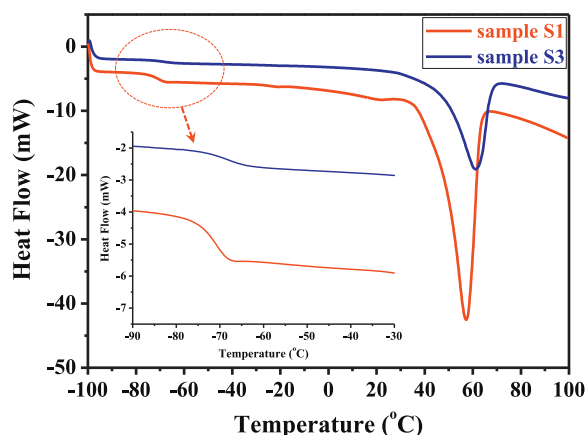


Fig. 4. Differential scanning calorimetry traces of sample S1 (PEO + LiPF₆) and sample S3 (PEO + MgAl₂O₄ + LiPF₆).

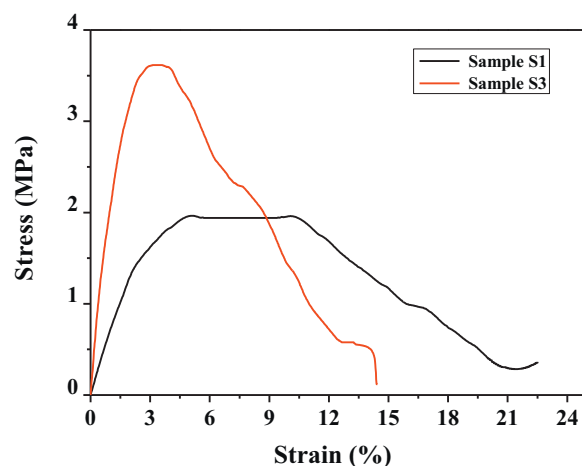


Fig. 5. Stress vs. strain behaviour of sample S1 (PEO + LiPF₆) and sample S3 (PEO + MgAl₂O₄ + LiPF₆).

identification of the extent of intercalation and exfoliation of the inert particles in the matrix. Fig. 4 depicts the DSC heating traces of samples S1 (PEO + LiPF₆) and S3 (PEO + LiPF₆ + MgAl₂O₄). Sample S1 shows the glass transition temperature (T_g) at -73 °C, whereas the MgAl₂O₄-incorporated sample S3 shows a slightly higher value of -70 °C. The increase in T_g of the composite polymer electrolyte might be due to (i) the effect of a small amount of dispersed particles and (ii) confinement of the intercalated/exfoliated polymer chains within the filler galleries, which resists the segmental motion of the polymer chains [33]. In particular, the increase in the T_g value of CPEs (towards the positive side) indicates better polymer–filler interaction, while the decrease in T_g (towards the negative side) indicates the plasticization effect. The marginal decrease in the melting point of the PEO polymer at higher concentration of fillers is attributed to the mild retarding effect on the crystallization that arises due to the added filler particles [34].

3.4. Tensile study

In order to quantify the mechanical strength, the stress–strain properties were measured. In general, the stress vs. strain behaviour shows two distinct regions: the initial linear region which reflects the elastic characteristic and the non-linear region which shows the plastic deformation. The stress vs. strain behaviour of samples S1 and S3 is shown in Fig. 5. It is evident from the figure that for sample S1, comprising 95% PEO + 5% LiPF₆, a brittle type of behaviour is seen as the film is not able to withstand the stress and fractures without yielding, thereby resulting in the absence of neck formation. Upon the addition of filler particles (10 wt.% of MgAl₂O₄) the ductility is improved, and sample S3 is able to withstand stress and the elastic deformation starts to localize, forming a neck before it fractures [35]. The tensile properties are presented in Table 2. The tensile strength of sample S1 (95% PEO + 5% LiPF₆) is 1.92 MPa (modulus of elasticity is 0.58 MPa) with an elongation-at-break value of 20.50%. Upon the

Table 2

Tensile properties of samples S1 (PEO + LiPF₆) and S3 (PEO + MgAl₂O₄ + LiPF₆).

Sample	Tensile strength (MPa)	Elongation at break (%)	Modulus of elasticity (MPa)
S1	1.92	20.50	0.58
S3	3.72	12.50	1.35

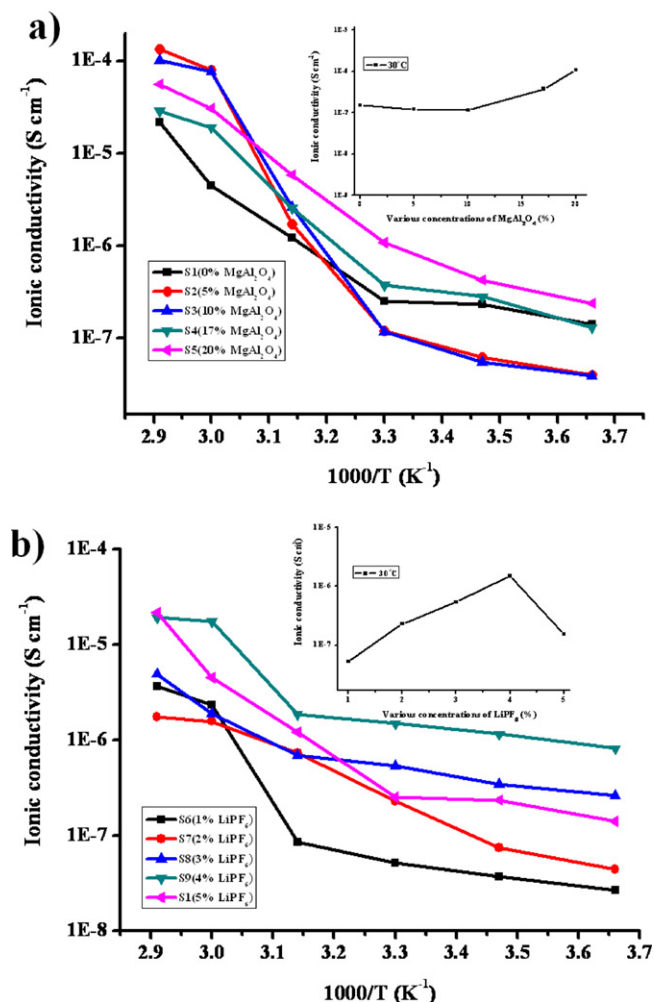


Fig. 6. Ionic conductivity as a function of $1/T$ for (a) various concentrations of filler (MgAl_2O_4) and (b) various concentrations of lithium salt, LiPF_6 . Inset shows the ionic conductivity at ambient temperature.

addition of MgAl_2O_4 , one can see that both modulus (1.35 MPa) and tensile strength increase (3.72 MPa) with dramatic decrease in strain at break 12.50% [36]. The maximum tensile strength (3.72 MPa) and moduli are obtained for sample S3 upon 10 wt.% addition of MgAl_2O_4 .

3.5. Ionic conductivity

In Fig. 6(a) and (b) the variation of the ionic conductivity as a function of the temperature is shown for different concentrations of lithium salt and filler, respectively. It is well known that Li ions migrate in two ways: (a) along the molecular chains of polymer, and (b) in the amorphous phase of polymer electrolyte [37,38]. Fig. 6(a) illustrates that the ionic conductivity increases with the increase of both temperature and filler concentration. The ionic conductivity gradually increases also with the increase in the LiPF_6 content up to 4% (samples S6–S8), as shown in Fig. 6(b). These results are in accordance with those reported on PEO-based polymer electrolytes with SiO_2 –lithium imide anion system [39]. The volume of the interfacial layers, which possesses a higher ionic conductivity, increases with the amount of MgAl_2O_4 filling in the polymer film leading to an enhancement in the number of tunnels for lithium ions migration. On the other hand, the competition between MgAl_2O_4 and hexafluorophosphate atoms with respect to lithium ions facilitates the migration for lithium ions in the polymer electrolyte. So, the ionic

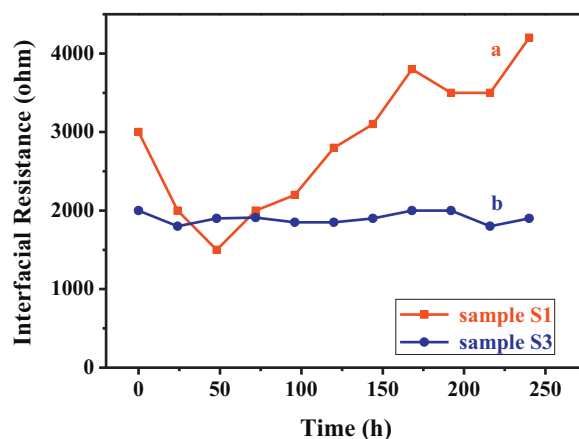


Fig. 7. Variation of interfacial resistance (R_i) as a function of time for the two symmetric cells: (a) Li/S1 ($\text{PEO} + \text{LiPF}_6$)/ Li and (b) Li/S3 ($\text{PEO} + \text{MgAl}_2\text{O}_4 + \text{LiPF}_6$)/ Li at 50 °C.

conductivity of the polymer electrolyte increases with the addition of MgAl_2O_4 into the polymer matrix.

According to Wicczorek et al. [40], the Lewis acid–base reactions between the filler surface and the PEO segments may induce structural modifications in the polymer matrix. The Lewis acid character of the added fillers would compete with the Lewis acid character of the lithium cations for the formation of complexes with the PEO chains which lowers the polymer chain reorganization tendency, thus promoting an overall stiffness to the structure. The resulting structure provides Li^+ -conducting pathways at the filler surface and enhances ionic transport.

3.6. Interfacial property

In present study, the compatibility of the composite polymer electrolyte comprising MgAl_2O_4 and LiPF_6 was examined by assembling Li/CPE/Li symmetric cells as described in Section 2.4. In order to employ the composite polymer electrolytes in the lithium polymer cells and to understand the role of the inert filler in the interfacial properties of the symmetric cells, samples S1 and S3 were used. Among the filler incorporated samples, sample S3 was selected from this test onwards, as it was found to be optimal in terms of ionic conductivity, thermal stability and mechanical integrity points of view. The variation of interfacial resistance as a function of time for the two symmetric cells composed of Li/CPE/Li at 50 °C is displayed in Fig. 7. It is clearly seen that the MgAl_2O_4 -incorporated CPE shows lesser interfacial resistance values than the electrolyte membrane prepared without filler; furthermore, the interfacial resistance remains stable over long periods of storage (240 h). The reduction in the interfacial resistance values of filler-added membrane is attributed to the formation of insulated layer of ceramic particles at the electrode surface. The inert particles depending upon the volume fraction would tend to minimize the area of the lithium electrode exposed to polymers containing O, OH-species and, thus, reduce the passivation process. It is also foreseeable that smaller size particles for a similar volume fraction of the ceramic phase would impart an improved performance as compared to larger size particles because they will cover more surface area [25]. The formation of an insulated layer of ceramic particles at the electrode surface is probable at a higher volume fraction of a passive ceramic phase. This insulating layer will impede electrode reactions. This may very well have happened when an excessive amount of the passive ceramic phase was introduced into the polymer matrix.

3.7. Lithium transference number

Although high ionic conductivity, enhanced thermal stability and good compatibility of lithium electrodes are desirable properties, they are not sufficient enough to make an electrolyte membrane useful for practical applications. The fraction of the current carried by Li^+ ions through the polymer electrolyte is fundamental for the performance of lithium batteries. To understand the effectiveness of MgAl_2O_4 in enhancing ion transport, transference number studies were carried out on samples S1 and S3 at 50°C , a temperature that is found to be optimal from a conductivity point of view for practical applications. The Li^+ transference numbers for samples S1 and S3 are 0.49 and 0.53, respectively. According to Capiglia et al. [39], who reported the transference numbers in the $\text{P}(\text{EO})_{20}\text{-LiBETI-SiO}_2$ system, increase in transference number for Li^+ ions is due to moisture trapped in the nanoparticles of SiO_2 . In the present study, since the CPE is prepared by a hot press technique with dry constituents, the increase in transference number may be attributed to increased salt dissociation due to the formation of ion-filler complexes by Lewis acid–base interactions between the surface groups of the MgAl_2O_4 filler and anions. Lithium ions are expected to move freely along these ceramic surface pathways; thus, under these conditions a consistent enhancement of cation transference number is logically expected. The polymeric membranes which have transference number above 0.5 are sufficient for lithium battery applications. In the present study, the S3 sample shows a t_{Li^+} value higher than 0.5 which renders it to be qualified for practical battery applications [41].

3.8. Charge–discharge studies

Fig. 8(a) and (b) shows the cycling (voltage/time) profile (plot a) and the charge/discharge capacity as a function of cycle number of $\text{Li/CPE (sample S3)/LiFePO}_4$ tested at different C-rates at 70°C . From the examples of charge–discharge profiles obtained between 2.5 and 4.0 V vs. Li/Li^+ at low C-rate, it can be seen that the cell shows a near 3.5 V potential plateau reflecting the common behaviour of the composite LiFePO_4/C cathode [42]. It is clear from plot b in Fig. 8 that the cell delivers an initial specific capacity approaching 160 mAh g^{-1} during its first 10 charge/discharge cycles at low C/10 rate. The capacity slightly reduces to about 150 mAh g^{-1} at C/5 up to the 25th cycle. During the first cycles at high 1 C current rate, the cell delivers a specific capacity of about 130 mAh g^{-1} up to 100 cycles, with a limited fade in capacity of about 0.3 mAh g^{-1} per cycle. In fact, the capacity retention is satisfactory: after 100 cycles, the cell still delivers specific capacity values around 110 mAh g^{-1} , with a drop in discharge capacity of less than 15% with respect to the initial cycle at the same current regime. A similar observation was reported by Appetecchi et al. [13], who reported the cycling behaviour in lithium cell of a $\text{PEO/LiCF}_3\text{SO}_3/\text{SiO}_2$ -based composite polymer electrolyte at 70°C . However, the present system shows better cycling stability and capacity retention than those reported earlier at a lower temperature [13,41–44]. Thus, the cell operates with the expected voltage profiles delivering a high fraction of the theoretical capacity. The cycle response is encouraging since only negligible decay in capacity is shown during this initial test and the Coulombic efficiency following the first cycles, where rearrangements in the structure of the electrode take place, is maintained to be always higher than 98%. These good performances are a convincing indication of a good interfacial contact between electrodes and separator, the efficient ionic conduction in the composite polymer separating electrolyte and the favourable interfacial charge transport between electrodes and electrolyte in the cell. These results warrant for an all-solid-state lithium-polymer cell conceived for safe application at elevated temperature.

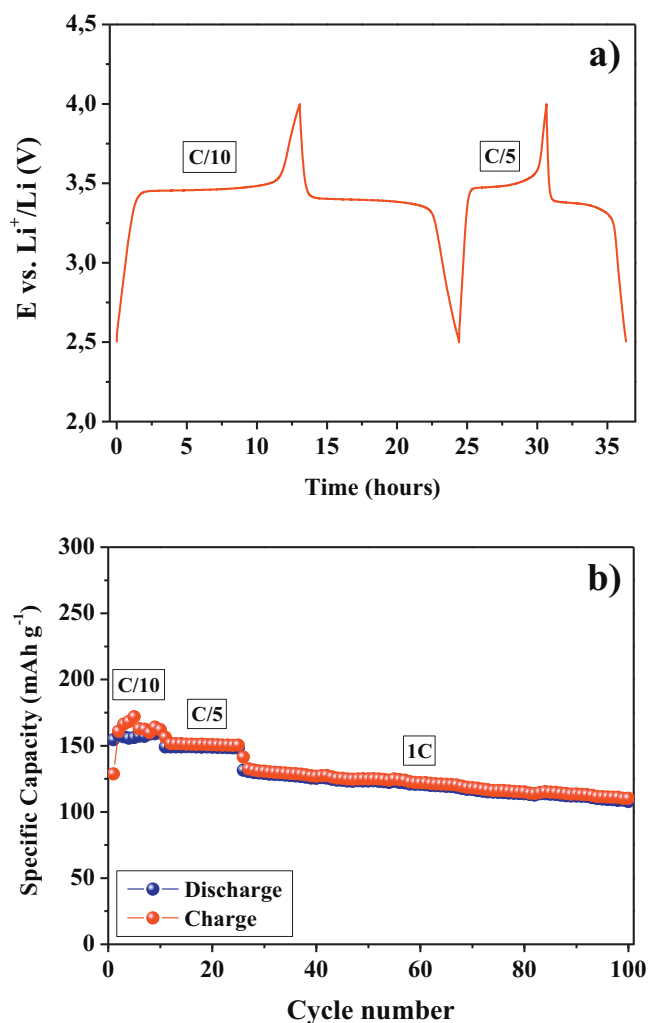


Fig. 8. (a) Cycling profile of $\text{Li/S3 (PEO + MgAl}_2\text{O}_4 + \text{LiPF}_6)/\text{LiFePO}_4$ cell at 70°C . (b) Charge and discharge capacity vs. cycle number of $\text{Li/S3 (PEO + MgAl}_2\text{O}_4 + \text{LiPF}_6)/\text{LiFePO}_4$ cell at 70°C .

4. Conclusions

In the present study, truly solid composite polymer electrolytes, comprising different concentrations of MgAl_2O_4 particles and LiPF_6 salt were prepared by a rapid hot press procedure and thoroughly studied. The composite polymer electrolyte membrane having the composition of $\text{PEO (85\%)} + \text{MgAl}_2\text{O}_4 \text{ (10\%)} + \text{LiPF}_6 \text{ (5\%)}$ was found to be optimum in terms of ionic conductivity and mechanical integrity. The all-solid-state lithium polymer cell assembled by sandwiching the above reported best performing composite electrolyte between a LiFePO_4 cathode and a lithium metal anode delivered a discharge capacity approaching 110 mAh g^{-1} after prolonged cycling (up to 100 cycles) at 70°C and high 1 C-rate which was found to be higher than that reported earlier for similar systems.

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